A VISUAL STUDY GUIDE

The Field Guide to Global Payments

How money actually moves — cards, bank rails, wallets, crypto, and the rules and business models built on top of them. The book's key points, in six pages, no cover-to-cover read required.

Sophia Goldberg • Stripe Press, 2022 • 4 sections, pp. 12–172

1 · CARDS	The four-party model, the transaction lifecycle, fees, declines & chargebacks.	pp. 12–72
2 · NON-CARD TAXONOMY	Bank, delayed, wallets, cash/ATM, crypto — what moves money when it's not a card.	pp. 73–122
3 · RULES & RISK	Regulation, compliance & fraud — the rulebook and the rule-breakers.	pp. 123–143
4 · BUSINESS MODELS	Subscriptions, marketplaces & B2B — how funds flow changes the business.	pp. 144–172

HOW TO USE THIS GUIDE

- Page 2 maps one card transaction end-to-end — the single diagram worth memorizing.
- Pages 3–5 unpack the three other sections as tables and diagrams you can scan in a minute each.
- Page 6 compares the three business models side-by-side and ends with the ten takeaways that matter most.

THE ONE-SENTENCE VERSION

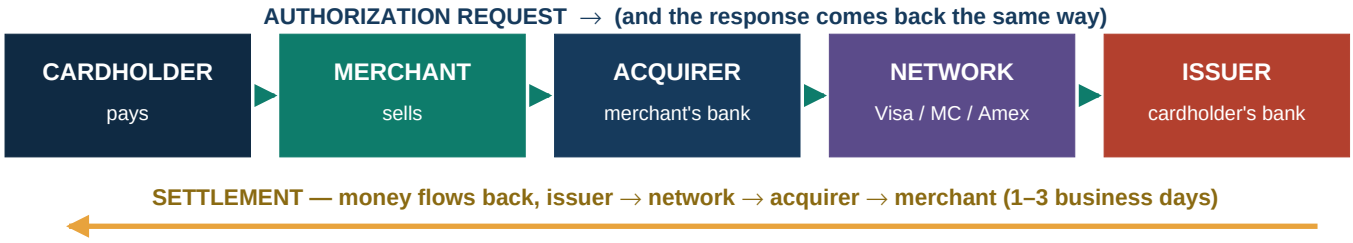
Every payment is the same question four ways: **who moves the money, who takes the risk, who sets the rules, and who gets paid for it.** Cards answer it one way; every other method answers it differently — and the business model decides which answer wins.

Prepared as a standalone visual study guide. All framing follows the book's four sections; figures (e.g. the ~85% authorization rate) are the book's own. Reference: fieldguidetoglobalpayments.com

SECTION 1 · CARDS — pp. 12–72

One card transaction, mapped

Four parties, one network in the middle, and two separate phases: authorization (a hold, in seconds) and clearing/settlement (the money, in days). Nothing is paid at the till — only promised.



The network is the clearing house: it routes messages, sets the rules (ISO 8583), and nets out who owes whom. Gateways & processors sit between merchant and acquirer; ISOs and PayFacs sell merchants access to all of it.

1 · AUTHORIZATION REQUEST	Merchant asks the issuer, via acquirer + network, to **hold** the funds. No money moves.
2 · AUTHORIZATION RESPONSE	Issuer approves or declines in 1–2 seconds. Approval = a hold on the cardholder's limit, not a payment.
3 · CAPTURE / CLEARING	Merchant batches approved sales at day's end; the network clears them — agreeing what each issuer owes.
4 · SETTLEMENT	Issuers pay the network, the network pays acquirers, acquirers deposit to the merchant — minus fees.

WHERE A $100 SALE GOES	
Interchange → issuer	Largest slice. Set by the network; varies by card type & region.
Scheme fee → network	The network's cut for routing, clearing & rules.
Acquirer markup	The acquirer's margin for the merchant account & risk.
Merchant keeps	≈ **$97–98 of every $100** after fees — before chargebacks.

DECLINES & LIABILITY — THE CALLOUTS

Only **~85% of authorizations succeed**. Top decline reasons: insufficient funds, expired card, fraud flags. Merchants that read decline codes and retry smartly recover real revenue.

Card-present vs card-not-present: in-store (chip/PIN) the issuer bears most fraud liability; online, without 3D Secure, the merchant usually eats it. 3D Secure shifts liability back — at the cost of checkout friction.

Chargeback: the cardholder disputes, the issuer claws the money back, and the merchant pays a fee on top — even when the dispute is friendly fraud or a service issue.

History in one line: "credit card" first appears in 1887 · Charg-It, the first bank card, 1946 · Diners Club & Amex, 1950s · BankAmericard launches the open loop in 1958 — and becomes Visa.

SECTION 1 · CARDS (continued)

Cards, unpacked

The four-party model is the grammar; everything else in the section is vocabulary — card types, loops, acceptance environments, and the intermediaries that sell access.

PLAYER	DOES WHAT	GETS PAID HOW
Cardholder	Buys; repays the issuer (credit) or spends own balance (debit).	Pays interest & fees (credit); nothing per-swipe (debit).
Merchant	Sells goods/services; accepts cards via an acquirer.	Receives sale value minus interchange, scheme & acquirer fees.
Issuer	Issues the card; approves/declines; holds the cardholder relationship.	Interchange — the largest fee slice — plus interest.
Acquirer	The merchant's bank; routes auth requests; settles funds to the merchant.	Acquirer markup on top of interchange + scheme fees.
Network	Visa / Mastercard / Amex. Routes messages, clears, sets the rules.	Scheme fees; also sets interchange levels.
Gateway / Processor	Tech layer: moves transaction data merchant ↔ acquirer.	Per-transaction or monthly tech fees.
ISO / PayFac	Sell & underwrite merchant accounts; PayFacs onboard sub-merchants fast.	Revenue share of processing; PayFacs take on compliance risk.

THREE CARD TYPES

Debit — pulls straight from checking; low fees, most common for everyday spend.

Credit — borrow up to a limit; issuer carries the credit risk (and earns interest).

Prepaid — pre-loaded funds; gifts, payroll, the unbanked.

OPEN VS CLOSED LOOP

Open loop (Visa, Mastercard): any merchant that accepts the network. Four parties.

Closed loop (Amex-issued, store cards): one issuer = the network. Fewer parties, tighter control.

Co-branded & dual-branded cards mix the two for rewards and reach.

WHERE YOU PAY MATTERS

Card-present (chip/PIN/tap): lower fraud, issuer bears most liability.

Card-not-present (online): higher fraud; liability usually sits with the merchant.

Customer-initiated vs merchant-initiated (recurring) changes who can retry — and who is liable.

ORCHESTRATION — THE QUIET UPGRADE

Modern gateways don't just pass messages: orchestration platforms **route each transaction across processors, retry declines intelligently, and give merchants one view of every rail.** Same four-party model underneath — smarter traffic control on top.

Cross-border is where cards get expensive: currency conversion, local regulation and correspondent banking (SWIFT) stack fees and delay on top of the normal flow — one reason non-card rails exist at all.

SECTION 2 · NON-CARD TAXONOMY — pp. 73–122

Everything that isn't a card

Cards dominate the US. Almost everywhere else, the default is something else — and each alternative has its own flow, its own refund rules, and its own failure modes. The book's point: there is no universal rail.

BANK	**ACH, wires, SEPA, instant schemes** (Pix, UPI, Faster Payments). Money moves account-to-account — no card network, no interchange.	Cheap per-transaction; slow (ACH) or fast (instant rails). **Watch-out:** settlement delays strain cash flow; refunds are clunky.
DELAYED	**Direct debit & BNPL.** Direct debit: payer authorizes the payee to pull funds on a schedule. BNPL: consumer pays over time, merchant is paid up front.	Predictable revenue, lower cost than cards. Guaranteed ACH cuts NSF risk. **Watch-out:** mandates, failed pulls, BNPL credit risk sits with the lender.
WALLETS	**Apple Pay, Google Pay, PayPal, Alipay, M-Pesa.** Stored funds or tokenized credentials; one tap, no card details shared with the merchant.	Cuts checkout friction and lifts conversion; tokens survive card reissue. **Watch-out:** wallet = a new gatekeeper in the flow.
CASH / ATM	**Cash, cash-on-delivery, ATM networks.** Still the norm in much of the world — Brazil treats cash installments as standard.	Zero chargebacks, universal acceptance. **Watch-out:** no digital trail, theft & handling cost, no refunds path.
CRYPTO / DeFi	**Stablecoins, on-chain settlement, DeFi rails.** Settlement without banks or card networks; programmable money.	Borderless and fast where rails are thin. **Watch-out:** volatility, regulation, and no chargeback mechanism at all.

THE REGIONAL MAP — SAME PURCHASE, DIFFERENT DEFAULT

United States	Cards dominate; wallets rising fast.
Germany	Debit & online bank transfer beat credit cards.
Japan	~65% of online transactions run through mobile wallets.
Brazil	Cash installments are normal; BNPL is booming.
China / India	Wallets & QR (Alipay, Paytm, UPI) are the default rails.
East Africa	Mobile money (M-Pesa) leapfrogged cards entirely.

Merchant's headache, per the book: every non-card method has **different refund and chargeback rules** — or none at all — so accounts receivable gets complicated the moment you accept more than one rail.

SECTION 3 · THE RULES AND THE RULE BREAKERS — pp. 123–143

Rules & risk cheat sheet

Payments run on rules written by networks, regulators and governments — and the fraud economy exists to find the gaps between them. Know the rule, know who it protects, know what breaks when it's ignored.

RULE / STANDARD	WHAT IT DOES	WHO IT HITS
PCI DSS	Security standard for handling card data: encryption, access control, audits. Non-compliance = fines & losing card acceptance.	Every merchant & processor touching card data.
Tokenization	Replaces real card numbers with tokens, so a breach yields useless data. Wallets run on it.	Merchants, gateways, wallets.
3D Secure (3DS)	Extra authentication step for online payments (SCA under PSD2). Shifts fraud liability to the issuer.	E-commerce merchants; adds checkout friction.
PSD2 / SCA	EU rule: strong customer authentication + open banking access for third parties.	EU merchants, banks, fintechs.
AML / KYC	Anti-money-laundering + know-your-customer checks on who moves money and why.	Banks, PayFacs, marketplaces, crypto on-ramps.
Network rules	Visa/Mastercard operating rules: chargeback windows, dispute categories, fine schedules.	Issuers, acquirers — and merchants, via their acquirer.
Cross-border rules	Local licensing, data residency, FX controls; SWIFT + correspondent banks enforce them.	Anyone selling across borders.

THE RULE-BREAKERS — FRAUD, BY TYPE

CARD-NOT-PRESENT FRAUD	Stolen card details used online, where there's no chip or PIN to check. The merchant usually bears the loss.
FRIENDLY FRAUD	A real cardholder disputes a real purchase — buyer's remorse, forgotten subscriptions, family spending. Merchant pays the chargeback + fee.
ACCOUNT TAKEOVER	Attacker logs in as the customer (credential stuffing, phishing) and spends stored payment methods. Wallets & tokens raise the bar.
FIRST-PARTY / SYNTHETIC	Fake identities built from real + invented data to open accounts and bust out. KYC and device signals are the defense.

THE TRADE-OFF THAT RUNS THE WHOLE SECTION

Every security control is a tax on conversion: **3DS cuts fraud and adds friction; tokenization costs integration effort and kills breach risk; strict KYC slows onboarding and stops bust-outs.** The book's lesson isn't “more rules” — it's knowing which trade you're making, and who pays for it.

SECTION 4 · BUSINESS MODELS — pp. 144–172

Subscriptions vs marketplaces vs B2B

Same rails, different money flow. The business model decides who holds funds, who bears the risk, and which payment features actually matter.

	SUBSCRIPTIONS	MARKETPLACES	B2B
Money flow	Recurring pulls from a stored credential — card-on-file, wallet token, or direct-debit mandate.	Buyer pays the platform; the platform splits & pays out many sellers (escrow in between).	Invoices paid by wire / ACH; large, lumpy, scheduled — not per-click.
What breaks	Involuntary churn: expired cards, failed pulls, dunning. Wallet tokens survive reissue; cards don't.	Splitting complexity, payout timing, seller fraud, and who holds funds (licensing!).	Slow settlement, FX on cross-border invoices, reconciliation across systems.
Key payments feature	Stored credentials, smart retries, dunning flows, direct-debit mandates for fixed amounts.	Split payouts, escrow, seller onboarding + KYC, instant vs batched payouts.	ACH/wire rails, virtual cards for AP, net terms, FX handling.
Who bears the risk	Merchant: churn + chargebacks on recurring billing.	Platform: it sits between buyer and seller money — and regulators notice.	Both sides: large tickets mean one failed payment is a real event.
Scale signal	Predictable revenue; retention is the metric.	Take rate × GMV; payout speed is the moat.	Larger transaction sizes than consumer payments, by an order of magnitude.

KEY TAKEAWAYS — THE TEN THAT MATTER

1. A card payment is a **promise**, not a payment — authorization holds funds; settlement moves them days later.
2. Four parties + one network: cardholder, merchant, issuer, acquirer — the network routes, clears and sets the rules.
3. Only **~85% of authorizations succeed**; declines are a revenue problem, not just a fraud problem.
4. Fees stack: interchange (issuer) + scheme (network) + acquirer markup — the merchant keeps ~$97–98 of $100.
5. Card-present vs card-not-present decides **who eats fraud**; 3DS shifts liability back to the issuer.
6. Outside the US, cards are the exception: wallets, bank rails and cash are the defaults — region picks the rail.
7. Every non-card method has different refund/chargeback rules — or none — which complicates accounts receivable.
8. Security is a trade: 3DS, tokenization and KYC each cut fraud and cost conversion. Know which trade you're making.
9. Subscriptions live or die on stored credentials; marketplaces on split payouts; B2B on large, slow, invoice rails.
10. Payments aren't a back office — they're where a business's revenue **literally comes from**.

Source: Sophia Goldberg, The Field Guide to Global Payments (Stripe Press, 2022) — fieldguidetoglobalpayments.com. Section page ranges per the book. Figures (e.g. ~85% authorization rate, ~65% wallet share in Japan) are the book's own. This guide is a study summary, not a substitute for the book.